

Microeconomics

Lecture 4: Individual Demand and Comparative Statics

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What We Know So Far

- **Budget constraint:** $p_x x + p_y y = M$
- **Optimal choice:** $MRS = p_x/p_y$ (tangency)
- **Solution methods:** Substitution and Lagrangian
- **Cobb-Douglas formula:** $x^* = \frac{\alpha}{\alpha+\beta} \cdot \frac{M}{p_x}$

Result: Given (p_x, p_y, M) we can find (x^*, y^*)

Next step

What happens to x^* when prices or income change? → We obtain the **demand function!**

Lecture Objectives

After this lecture you will be able to:

1. **Derive** the individual demand function from the optimal choice
2. **Graph** the demand curve
3. **Distinguish** normal goods from inferior goods (Engel curve)
4. **Understand** the price effect and its decomposition
5. **Analyze** the substitution effect and the income effect (introduction)
6. **Recognize** Giffen goods (the demand paradox)

Lecture Focus

Understand how quantity demanded changes as prices and income change, and why.

Deriving Demand: From Optimal Choice to Demand

Optimal choice: For each triple (p_x, p_y, M) , we find the optimal (x^*, y^*) .

Demand function:

If I vary one parameter (e.g., p_x), holding the others fixed (p_y, M), I obtain the relationship:

$$x^* = x^D(p_x; p_y, M)$$

This is the (individual) **demand function** for good x .

Definition: Marshallian Demand

The **Marshallian demand function** gives the optimal quantity demanded as a function of prices and income:

$$x^D = x^D(p_x, p_y, M)$$

$$y^D = y^D(p_x, p_y, M)$$

Example: Cobb-Douglas

Utility function: $U(x, y) = x^\alpha y^\beta$

Optimal choice:

$$x^* = \frac{\alpha}{\alpha + \beta} \cdot \frac{M}{p_x}$$
$$y^* = \frac{\beta}{\alpha + \beta} \cdot \frac{M}{p_y}$$

These are already the **Marshallian demand functions!**

Demand function for x :

$$x^D(p_x, p_y, M) = \frac{\alpha}{\alpha + \beta} \cdot \frac{M}{p_x}$$

Similarly for y :

$$y^D(p_x, p_y, M) = \frac{\beta}{\alpha + \beta} \cdot \frac{M}{p_y}$$

Note (important): With Cobb-Douglas the expenditure share on x is constant and equal to $\alpha/(\alpha + \beta)$, so x^D *does not depend* on p_y (a structural property, not a cancellation).

Numerical Example

Data: $U = x^{0.5}y^{0.5}$ ($\alpha = \beta = 0.5$), $p_y = 5$, $M = 100$

Demand function:

$$x^D(p_x) = \frac{0.5}{1} \cdot \frac{100}{p_x} = \frac{50}{p_x}$$

Table: Quantity demanded at various prices

p_x	x^D
2	25
5	10
10	5
20	2.5
25	2

Remark: As p_x falls, x^D rises (inverse relationship).

Demand Curve: Definition

Demand Curve

The **demand curve** is the graphical representation of the relationship between price and quantity demanded, *holding other factors fixed* (p_y , M).

Graphical convention:

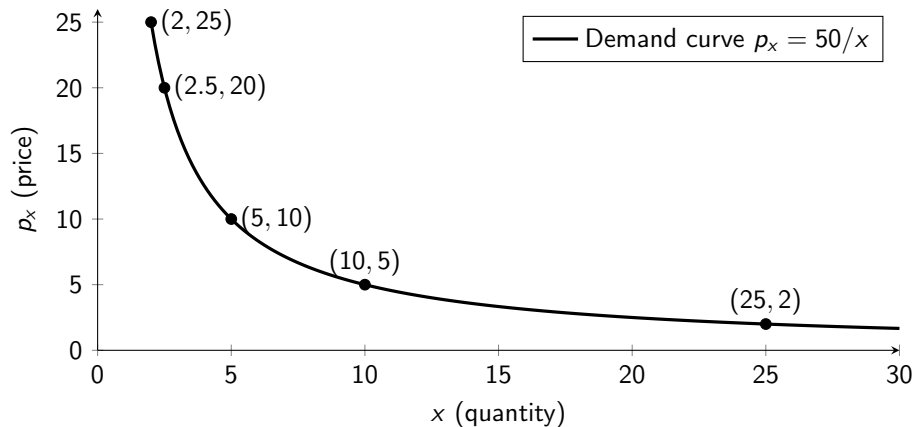
- **Horizontal axis:** Quantity x (dependent variable)
- **Vertical axis:** Price p_x (independent variable)

Historical note: “Reversed” convention introduced by Marshall—standard in economics.

For our example: $x^D = 50/p_x \rightarrow$ inverse form: $p_x = 50/x$

Demand Curve: Graph

Example: $x^D = 50/p_x$ (equivalent to $p_x = 50/x$)



Property: Negative slope $\rightarrow$ **law of demand.**

Properties of Marshallian Demand

1. Homogeneity of Degree Zero

If I multiply all prices and income by $\lambda > 0$:

$$x^D(\lambda p_x, \lambda p_y, \lambda M) = x^D(p_x, p_y, M)$$

Interpretation: Only *relative* prices/income matter.

Check for Cobb-Douglas:

$$x^D(\lambda p_x, \lambda p_y, \lambda M) = \frac{\alpha}{\alpha + \beta} \cdot \frac{\lambda M}{\lambda p_x} = \frac{\alpha}{\alpha + \beta} \cdot \frac{M}{p_x} \quad \checkmark$$

2. Negative Slope (typical)

For most goods:

$$\frac{\partial x^D}{\partial p_x} < 0$$

Note: There are rare exceptions (Giffen goods)

Shifts of and Movements along the Curve

Fundamental distinction:

1. Movement **ALONG** the curve

- Cause: change in p_x (the good's own price)
- Effect: we move from one point to another on the same curve

2. Shift **OF** the curve

- Cause: change in M (income) or in p_y (price of other goods)
- Effect: the whole curve shifts (right/left)

Rule:

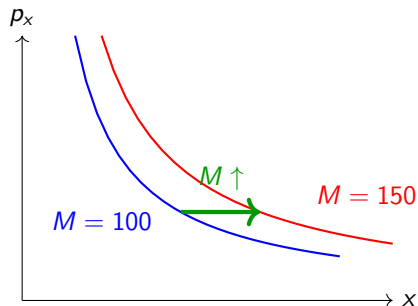
- p_x changes $\rightarrow$ movement along
- M or p_y changes $\rightarrow$ shift of the curve

Example: Shift in Demand

Function: $x^D = \frac{M}{2p_x}$ (Cobb-Douglas with $\alpha = \beta$)

Scenario 1: $M = 100 \rightarrow x^D = 50/p_x$

Scenario 2: M rises to 150 $\rightarrow x^D = 75/p_x$



Interpretation: With more income, at every price the consumer demands *more* $x \rightarrow$ the curve shifts right.

Price of Other Goods: Substitutes and Complements

What happens if the price of the other good y changes?

Effect of the price of the other good (cross-price)

Look at the sign of:

$$\frac{\partial x^D}{\partial p_y}$$

- **Substitutes:** if $p_y \uparrow \Rightarrow$ I buy more $x \Rightarrow \frac{\partial x^D}{\partial p_y} > 0$
- **Complements:** if $p_y \uparrow \Rightarrow$ I buy less $x \Rightarrow \frac{\partial x^D}{\partial p_y} < 0$

Key idea

A change in p_y does not make you “move along” the demand for x : it **shifts** the demand curve for x .

Exercise: Substitutes or Complements?

Two goods: **pizza** (x) and **beer** (y). The demand for pizza is:

$$x^D(p_x, p_y, M) = 20 - 2p_x + p_y + 0.01M$$

Questions:

1. Is the sign of $\frac{\partial x^D}{\partial p_y}$ positive or negative?
2. Are pizza and beer **substitutes** or **complements**?
3. If p_y rises by 2 euros (from 3 to 5), by how much does x^D change?

Solution:

- $\frac{\partial x^D}{\partial p_y} = +1 > 0$
- Hence **substitutes**: if beer becomes more expensive, I buy more pizza
- If p_y rises by 2: $\Delta x^D = 1 \cdot 2 = +2$

Derivation from Other Types of Preferences

Perfect substitutes: $U = ax + by$

- If $a/b > p_x/p_y$: $x^D = M/p_x$ (I buy only x)
- If $a/b < p_x/p_y$: $x^D = 0$ (I buy only y)
- If $a/b = p_x/p_y$: x^D indeterminate (any combination)

→ Demand is **discontinuous** at the critical price.

Perfect complements: $U = \min\{ax, by\}$

- Optimum: $ax^* = by^*$ always
- Solving with the constraint: $x^D = \frac{bM}{bp_x + ap_y}$

→ Demand is **continuous**, decreasing in p_x .

Exercise: Deriving Demand

Function: $U = x^{0.3}y^{0.7}$, $p_y = 4$, $M = 120$

Question a: Derive the demand function $x^D(p_x)$.

Solution:

$$x^D(p_x) = \frac{0.3}{0.3 + 0.7} \cdot \frac{120}{p_x} = 0.3 \cdot \frac{120}{p_x} = \frac{36}{p_x}$$

Question b: Compute x^D for $p_x = 3$ and $p_x = 6$.

Solution:

- $p_x = 3$: $x^D = 36/3 = 12$
- $p_x = 6$: $x^D = 36/6 = 6$

Question c: If M doubles, what happens to x^D ?

Solution: $x^D = \frac{0.3 \cdot 240}{p_x} = \frac{72}{p_x}$

→ Demand doubles at every price (the curve shifts to the right) ✓

From Individual Demand to Market Demand

So far: we have derived **individual demand**:

$$x_i^D = x_i^D(p_x, p_y, M_i)$$

Market Demand

Market demand is the sum of individual demands:

$$X^D(p_x, p_y, \{M_i\}) = \sum_{i=1}^N x_i^D(p_x, p_y, M_i)$$

Interpretation:

- At each price p_x I add up the quantities demanded by all consumers
- The market demand curve is the **horizontal sum** of the individual curves

Example (2 consumers, Cobb-Douglas with $\alpha = \beta$):

$$x_1^D = \frac{M_1}{2p_x}, \quad x_2^D = \frac{M_2}{2p_x} \Rightarrow X^D = \frac{M_1 + M_2}{2p_x}$$

Exercise: Market Demand

Two consumers have individual demands:

$$x_1^D(p_x) = \frac{60}{p_x}, \quad x_2^D(p_x) = \frac{40}{p_x}$$

a) Find the market demand $X^D(p_x)$.

Solution:

$$X^D(p_x) = x_1^D + x_2^D = \frac{60}{p_x} + \frac{40}{p_x} = \frac{100}{p_x}$$

b) Compute X^D for $p_x = 5$ and $p_x = 20$.

Solution:

- $p_x = 5 \Rightarrow X^D = 100/5 = 20$
- $p_x = 20 \Rightarrow X^D = 100/20 = 5$

Summary of Part 1

1. Marshallian Demand Function

- $x^D(p_x, p_y, M)$ = optimal quantity as a function of the parameters
- Derived from the optimality condition

2. Demand Curve

- Graph: p_x vs x (typically decreasing)
- Movement along vs shift of the curve
- Cross-price: substitutes vs complements

3. Market demand

- Horizontal sum of individual demands

Engel Curves: Demand and Income

So far we have seen how x^D depends on p_x (holding M fixed).

Now we look at what happens if **income** M changes (holding prices fixed).

Engel Curve

The **Engel curve** shows the relationship between income and quantity demanded, at constant prices:

$$x^D = x^D(M; p_x, p_y)$$

Graph: M on the vertical axis, x on the horizontal axis.

Normal Goods and Inferior Goods

Classification based on how x^D responds to changes in M :

Normal Good

A good is **normal** if quantity demanded rises as income rises:

$$\frac{\partial x^D}{\partial M} > 0$$

Inferior Good

A good is **inferior** if quantity demanded falls as income rises:

$$\frac{\partial x^D}{\partial M} < 0$$

Examples:

- **Normal:** Quality food, designer clothes, travel
- **Inferior:** Cheap pasta, public transport (for some), low-quality housing

Example: Cobb-Douglas (Normal Good)

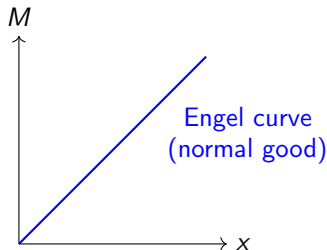
Demand function: $x^D = \frac{\alpha}{\alpha+\beta} \cdot \frac{M}{p_x}$

Derivative with respect to M :

$$\frac{\partial x^D}{\partial M} = \frac{\alpha}{\alpha + \beta} \cdot \frac{1}{p_x} > 0 \quad \text{always!}$$

Conclusion: With Cobb-Douglas preferences, all goods are **normal**.

Engel curve: Rewrite: $M = \frac{\alpha+\beta}{\alpha} \cdot p_x \cdot x$ (increasing straight line)



Limiting case of unit elasticity ($\eta_{x,M} = 1$): constant expenditure share, neither luxury nor necessity.

Luxury Goods and Necessities

A further distinction among normal goods:

Luxury (or Superior) Good

A normal good is a **luxury** if its share of income rises as M rises:

$$\frac{\partial(p_x x / M)}{\partial M} > 0$$

Necessity

A normal good is a **necessity** if its share of income falls as M rises:

$$\frac{\partial(p_x x / M)}{\partial M} < 0$$

Examples:

- **Luxuries:** Exotic travel, jewelry, starred restaurants
- **Necessities:** Bread, electricity, basic rent

Engel's Law (Empirical)

Engel's Law

As a household's income rises, the share of income spent on **food** falls (even though absolute spending rises).

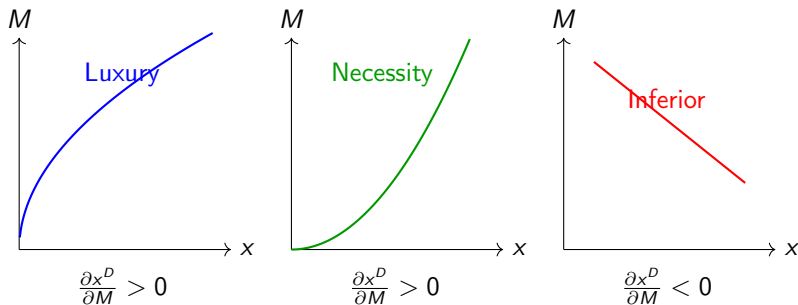
Modern interpretation:

- Food is a **normal** good
- But it is a **necessity** (the share $p_x x / M$ falls with M)

Modern empirical evidence (indicative):

- Poor households: higher share of income spent on food
- Rich households: lower share of income spent on food

Graph: Three Types of Engel Curves



Features:

- **Luxury:** concave curve (x grows more than proportionally in M , elasticity > 1)
- **Necessity:** convex curve (x grows less than proportionally in M , elasticity $\in (0, 1)$)
- **Inferior:** decreasing curve

Numerical Example: Inferior Good

Realistic case: Cheap pasta vs quality pasta

Scenario:

- With low income ($M = 1,000$ €/month): I buy 20 kg of cheap pasta
- With high income ($M = 3,000$ €/month): I buy 8 kg of cheap pasta (and more quality pasta/other foods)

Hypothetical demand function:

$$x_{\text{cheap pasta}}^D = 26 - 0.006M$$

Check:

- $M = 1,000$: $x^D = 26 - 6 = 20$ kg ✓
- $M = 3,000$: $x^D = 26 - 18 = 8$ kg ✓
- $\frac{\partial x^D}{\partial M} = -0.006 < 0 \rightarrow$ inferior good ✓

Interpretation: With more income, I replace cheap pasta with better alternatives.

Exercise: Identify the Type of Good

Hypothetical demand functions. Classify each good:

a) $x^D = 0.1M/p_x$

Answer: $\frac{\partial x^D}{\partial M} = 0.1/p_x > 0 \rightarrow$ **Normal**

Expenditure share: $(p_x x)/M = 0.1$ (constant) $\rightarrow$ unit elasticity: **neither luxury nor necessity** ✓

b) $x^D = 100 - 0.02M$

Answer: $\frac{\partial x^D}{\partial M} = -0.02 < 0 \rightarrow$ **Inferior** ✓

c) $x^D = 0.001M^2/p_x$

Answer: $\frac{\partial x^D}{\partial M} = 0.002M/p_x > 0 \rightarrow$ **Normal**

Expenditure share: $(p_x x)/M = 0.001M$ (rises with M) $\rightarrow$ **Luxury** ✓

d) $x^D = 5\sqrt{M}/p_x$

Answer: $\frac{\partial x^D}{\partial M} = \frac{5}{2\sqrt{M}p_x} > 0 \rightarrow$ **Normal**

Share: $(p_x x)/M = 5/\sqrt{M}$ (falls with M) $\rightarrow$ **Necessity** ✓

Summary of Part 2

1. Engel Curve

- Relationship between income M and quantity demanded x^D
- Graph: M vs x (fixed prices)

2. Classification of Goods

- **Normal:** $\partial x^D / \partial M > 0$ (rises with income)
- **Inferior:** $\partial x^D / \partial M < 0$ (falls with income)
- **Luxury:** Normal + expenditure share rises with M
- **Necessity:** Normal + expenditure share falls with M

3. Engel's Law

- Food expenditure share falls with income (necessity)
- Confirmed empirically

Decomposing the Price Effect

Question: When p_x changes, why does x^D change?

Answer: There are **two simultaneous effects**:

Substitution Effect

Change in quantity demanded due to the good having become relatively more or less expensive compared with other goods.

Direction: Always opposite to the price change.

Income Effect

Change in quantity demanded due to the change in the consumer's real purchasing power.

Direction: Depends on whether the good is normal or inferior.

Intuitive Example

Scenario: The price of pizza (p_x) rises from €5 to €10.

Substitution Effect:

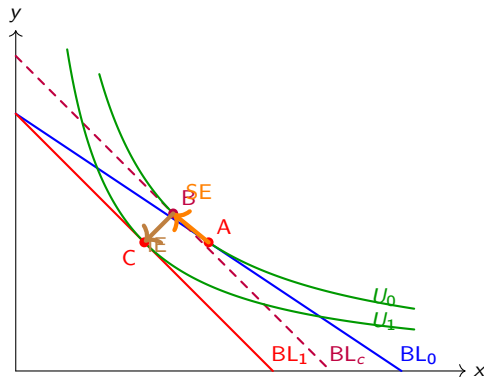
- Pizza relatively more expensive $\rightarrow$ I buy less pizza

Income Effect:

- Purchasing power falls
- If pizza is **normal**: I buy less pizza
- If pizza were **inferior**: I might buy more of it

Graph: Decomposition (Normal Good)

Standard case: p_x rises, normal good



SE: $A \rightarrow B$ (constant utility U_0 , substitution only)

IE: $B \rightarrow C$ (real income falls, utility drops to U_1)

$A \rightarrow B$ (SE): change in relative prices ($p_x/p_y \uparrow$) at constant utility $\Rightarrow$ I substitute y for $x \Rightarrow x$ falls.

$B \rightarrow C$ (IE): once the compensation is removed, real income falls; with x a normal good $\Rightarrow x$ falls further.

Total ($A \rightarrow C$): SE and IE go in the same direction $\Rightarrow x^D$ certainly falls.

Price-Consumption Curve

Definition

The **price-consumption curve** (PCC) is the locus of all optimal bundles as p_x varies (with p_y and M fixed).

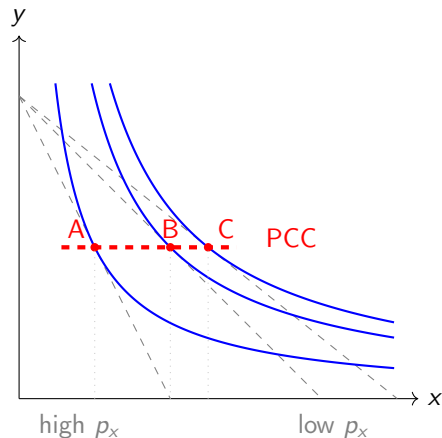
Graphical construction:

1. Fix (p_y, M)
2. Vary p_x : $p_x^1 < p_x^2 < p_x^3$
3. For each p_x^i : find the optimum (x_i^*, y_i^*) (tangency)
4. Connect all the optimal points $\rightarrow$ price-consumption curve

Interpretation:

- The PCC shows how the optimal choice changes when the price changes
- It is the “path” the consumer follows in the (x, y) space
- From the PCC we can derive the demand curve (projecting onto (x, p_x))

Graph: Price-Consumption Curve



Note: As p_x falls, the constraint rotates outward and the optimum moves along the PCC: x^* rises while y^* stays constant (in this Cobb–Douglas example with $\alpha = \beta$).

From the PCC to the Demand Curve

Key idea: each optimal point on the PCC corresponds to a different price p_x .

Construction (step by step)

1. Choose a price $p_x^i \Rightarrow$ find the optimum (x_i^*, y_i^*)
2. Associate the pair (x_i^*, p_x^i)
3. Repeat and connect the points

Conclusion: demand is the **projection** of the PCC onto the (x, p_x) plane.

The Giffen Good Paradox

Law of demand: $p_x \uparrow \rightarrow x^D \downarrow$ (negative slope)

But there is a rare theoretical exception:

Giffen Good

A **Giffen good** is a good whose quantity demanded *rises* when its price rises:

$$\frac{\partial x^D}{\partial p_x} > 0 \quad (\text{positive slope!})$$

Necessary conditions:

1. The good must be (strongly) **inferior**
2. The good must account for a **large** share of the budget
3. The income effect must dominate the substitution effect

Name: Robert Giffen (1837-1910), Scottish economist who observed this behavior in contexts of extreme poverty.

Empirical Evidence: Giffen Goods

Do they really exist?

Most cited cases:

- **Bread (England, 19th century)**: often cited, but historical data are limited
- **Potatoes (Ireland, 1845–1852)**: **controversial** case (famine = supply shock + physical scarcity)

Stronger evidence

China (Jensen & Miller, 2008): experiment on the price of the staple food
→ causal evidence of Giffen-like behavior among very poor households.

Why so rare?

- Extreme conditions are needed: poverty + strongly inferior good + huge budget share
- If the budget share is small, the income effect is too weak to dominate the substitution effect

Historical Example 1: Bread in Victorian England

Context: London, mid-1800s. Working-class families in extreme poverty.

Marshall's observation:

When the price of bread rises, some poor families *increase* their bread consumption.

Mechanism:

1. **Initial situation:**

- ▶ Bread = 70-80% of the household budget
- ▶ The rest: a little meat, butter, vegetables

2. **The price of bread rises:**

- ▶ Budget “eaten up” by the bread price increase
- ▶ They can no longer afford meat and butter
- ▶ They must cut the “extras” and buy **only** bread

3. **Paradoxical result:**

- ▶ Total bread consumption *rises*
- ▶ Bread becomes the only affordable source of calories

Historical Example 1: Bread (Economic Analysis)

Why might bread be a Giffen good?

Conditions satisfied

1. **Inferior good:** With more income, families would buy meat instead of bread alone
2. **Huge share of the budget:** 70-80% of income spent on bread
3. **Few alternatives:** In extreme poverty, there are no cheaper substitutes for bread as a source of calories

Decomposition of the effects:

- **Substitution effect:** Less bread (it is more expensive) → small
- **Income effect:** Much poorer → I must cut meat/butter → I buy more bread (the only affordable food) → **huge**
- **Total effect:** Bread consumption rises

Note: This example is cited in classic textbooks, but precise historical data are limited.

Historical Example 2: Potatoes in Ireland—The Traditional Reasoning

Context: The Great Famine in Ireland. Potato blight destroys the harvests.

Scenario:

- Potatoes: an **inferior** good (the poor eat them, the rich do not)
- Potatoes: a **huge** share of poor households' budgets (50-80%)
- The price of potatoes rises dramatically (famine)

Paradoxical reaction:

1. **Substitution effect:** $p_{\text{potatoes}} \uparrow \rightarrow$ I would like to substitute bread/meat
 $\rightarrow$ I buy fewer potatoes (standard effect)
2. **Income effect:** $p_{\text{potatoes}} \uparrow \rightarrow$ purchasing power collapses
 $\rightarrow$ Now much poorer, I must give up bread/meat (expensive)
 $\rightarrow$ I buy even more potatoes (the only affordable food!)
 $\rightarrow$ Huge and opposite income effect
3. **Total effect:** Income effect $>$ Substitution effect
 $\rightarrow x_{\text{potatoes}}^D$ rises! (paradox)

Potatoes in Ireland—Comparison with the Historical Evidence

The reasoning on the previous slide is the standard textbook one. But the historical evidence contradicts it.

Apparently Giffen scenario:

- Potatoes = staple food of the poor Irish population (50-80% of calories)
- The price of potatoes rises dramatically (scarcity)
- Potatoes are an inferior good (the rich do not eat them)

But the historical evidence says otherwise!

Modern studies of famine data show that:

- Potato consumption **fell** when prices rose
- People could not afford even potatoes (too expensive)
- Many starved to death or emigrated
- It is **not** a true example of a Giffen good

Lesson: Even when the theoretical conditions seem to be met, empirical evidence is crucial.

Why Ireland Is NOT a Giffen Example

Theoretical problem:

A Giffen good requires that the consumer can *choose* to buy more of the good when its price rises.

In the Irish famine:

1. **Physical scarcity:** There were not enough potatoes (destroyed by the blight)
 - ▶ Even if they wanted to, people could not buy more potatoes
 - ▶ It is a **supply** problem, not a demand problem
2. **Extreme poverty:** The price became so high that even potatoes became unaffordable
 - ▶ Families could not afford even the “inferior good”
 - ▶ People dropped out of the market altogether (hunger, death, emigration)
3. **Historical data:** The records show a **decrease** in potato consumption

Conclusion: The Irish example is often cited incorrectly. It illustrates a humanitarian tragedy, but not the Giffen paradox in the strict sense.

Modern Example: The Jensen & Miller (2008) Study in China

The first true controlled experiment on Giffen goods!

Context: Poor provinces of China (Hunan and Gansu), very low-income households.

Experimental design:

1. They identify the local “staple food”:
 - ▶ **Hunan:** Rice (50-60% of calories)
 - ▶ **Gansu:** Wheat/noodles (40-50% of calories)
2. They provide **randomized subsidies** on the price of the staple food
 - ▶ Treatment group: price artificially reduced
 - ▶ Control group: normal price
3. They measure the response of demand

The Jensen & Miller Study: Results

Evidence of Giffen behavior

When the price of the staple food **falls** (thanks to the subsidy):

- The poorest households **reduce** their consumption of the staple
- They switch to higher-quality foods (meat, vegetables)

Equivalently: when the price rises, they consume **more** of the staple

→ **Giffen behavior documented** (strong evidence for rice in Hunan, weak for wheat in Gansu)

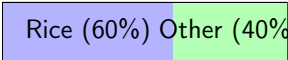
Why does it work here and not in Ireland?

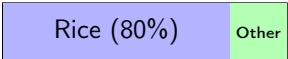
1. **Controlled conditions:** An experiment, not a famine
2. **Moderate changes:** Manageable price changes (no extreme shocks)
3. **Available supply:** The foods were available on the market
4. **Precise measurement:** Data collected systematically

Limitations: Small effect, only for very poor households and staples with a huge budget share.

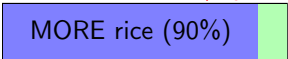
Mechanism of the Chinese Study (Visualization)

Budget of a poor household in Hunan:

Initial budget:  Rice (60%) Other (40%)

$p_{\text{rice}} \uparrow$:  Rice (80%) Other


I must cut meat/vegetables

Result:  MORE rice (90%)


I buy more rice (only source of calories)

Key: The income effect (increased poverty) forces consumption of the inferior good up by more than the substitution effect reduces it.

Note: the bars show the expenditure share. The move from 60%→80% is partly share arithmetic; the Giffen effect shows up in the jump to 90%, where the physical quantity of rice increases.

Summary: Empirical Evidence on Giffen Goods

Example	Status	Note
Bread (England, 1800s)	Possible, but limited data	Cited by Marshall; theoretical conditions present
Potatoes (Ireland, 1845-52)	NOT Giffen	Physical scarcity + extreme poverty; consumption fell
Rice/Wheat (China, 2008)	Confirmed (rice, Hunan)	First controlled experiment; weak evidence for wheat (Gansu)

General lessons:

1. Giffen goods **exist**, but they are extremely rare
2. Very specific conditions are needed: extreme poverty + inferior good + huge budget share
3. In developed countries they are practically nonexistent (welfare, variety of goods)
4. Empirical evidence is crucial: theoretical plausibility is not enough
5. Giffen goods are theoretically important for understanding the SE/IE decomposition, even if rare

Why Study Giffen Goods If They Are So Rare?

Three fundamental reasons:

1. Theoretical completeness

- ▶ They show that the “law of demand” is not universal
- ▶ There are logically possible exceptions
- ▶ Theory must also explain limiting cases

2. Understanding the mechanisms

- ▶ An extreme case that sheds light on the SE/IE decomposition
- ▶ Understanding Giffen goods → better understanding of substitution and income effects
- ▶ Useful also for “standard” normal and inferior goods

3. Relevance for poverty

- ▶ They show economic behavior under extreme poverty
- ▶ Implications for development policy (food subsidies)
- ▶ Jensen & Miller study: subsidies can have counterintuitive effects

Conclusion: Extreme cases help us better understand normal cases!

Synthesis: Price Effect by Type of Good

When p_x rises:

Type of Good	Substitution Effect	Income Effect
Normal	$x^D \downarrow$	$x^D \downarrow$
Inferior (typical)	$x^D \downarrow$	$x^D \uparrow$ (small)
Giffen (rare)	$x^D \downarrow$	$x^D \uparrow\uparrow$ (large)

Conclusion:

- For **normal** goods: The effects add up $\rightarrow$ demand always decreasing
- For **standard inferior** goods: Substitution effect dominates $\rightarrow$ demand decreasing
- For **Giffen** goods: Income effect dominates $\rightarrow$ demand increasing (extremely rare!)

Exercise 1: Qualitative Decomposition (1)

Scenario: The price of a (normal) good x goes from €5 to €8. Income M and the price of y stay constant. The quantity demanded of x goes from 10 to 6 units.

a) Explain in words how the **substitution effect** works.

Solution: Good x has become relatively more expensive than y . Even if I could keep the same utility level, I would prefer to substitute y for x because the price ratio has changed. The substitution effect always leads to a **reduction** in the quantity of x when p_x rises.

b) Explain in words how the **income effect** works.

Solution: With a higher p_x , my real purchasing power has fallen (it is as if I had become poorer). Since x is a **normal** good, when real income falls, I demand less x . The income effect therefore reinforces the substitution effect.

Exercise 1: Qualitative Decomposition (2)

c) Do the two effects go in the same direction or in opposite directions? Why, for a normal good, is the total result always “I demand less x ”?

Solution: The two effects go in the **same direction** (both reduce x^D):

- **Substitution effect:** $p_x \uparrow \rightarrow x$ relatively more expensive $\rightarrow x^D \downarrow$
- **Income effect:** $p_x \uparrow \rightarrow$ purchasing power $\downarrow \rightarrow$ (normal good) $\rightarrow x^D \downarrow$

Conclusion: For a normal good, both effects push in the same direction (a reduction in x^D), so the demand curve is **always decreasing**. There is no ambiguity.

General rule

Normal good + price increase = demand certainly falls (SE and IE agree)

Exercise 2: Inferior Good vs Giffen (1)

Scenario: “Cheap pasta” is an inferior good that accounts for 10% of a household’s budget. Its price rises by 10%.

a) Which way does the substitution effect go (more or less pasta)?

Solution: The substitution effect always leads to consuming **less** of the good that has become more expensive. Hence: less cheap pasta (I substitute other foods, even if only marginally).

b) Which way does the income effect go, given that it is an inferior good?

Solution: The price rises → purchasing power falls. Since cheap pasta is an **inferior** good, when I become poorer I demand **more** of it. The income effect therefore pushes toward **increasing** consumption of cheap pasta.

Note: The effects go in opposite directions!

Exercise 2: Inferior Good vs Giffen (2)

c) Is it plausible that cheap pasta is a Giffen good in this case? Explain why, reasoning about the “size” of the income effect.

Solution: No, it is not plausible.

- Cheap pasta accounts for only **10% of the budget**
- A 10% price increase reduces overall purchasing power by only **1%** (= 10% of 10%)
- The **income effect** is therefore very small
- The **substitution effect**, instead, acts directly on the relative cost of pasta
- Result: SE (negative) **dominates** IE (positive but small)
- The quantity demanded of cheap pasta **falls** anyway

Condition for Giffen

Goods must weigh **enormously** on the budget (50-80%) for the income effect to be strong enough to dominate the substitution effect.

Exercise 3: A Story to Classify (1)

Scenario: A poor worker spends 80% of his income on the “staple food” (rice), and the rest on meat and vegetables. When the price of rice rises sharply:

- He can hardly afford meat and vegetables any more
- He increases his rice consumption so as not to cut total calories too much

a) Is rice a normal or an inferior good?

Solution: Rice is an **inferior** good. If the worker had more income, he would reduce his rice consumption and increase that of meat/vegetables (higher-quality foods).

b) Does the substitution effect lead to more or less rice?

Solution: The substitution effect leads to consuming **less rice**. Rice has become relatively more expensive than meat and vegetables, so I would like to replace it with other foods.

Exercise 3: A Story to Classify (2)

c) Does the income effect lead to more or less rice?

Solution: The income effect leads to consuming **more rice**.

- The price of rice rises → purchasing power collapses dramatically (rice = 80% of the budget!)
- The worker becomes much poorer in real terms
- Since rice is **inferior**, when he becomes poorer he demands more of it
- He can no longer afford meat and vegetables (too expensive), so he must “fall back” on rice

d) Which effect must be larger for rice to behave like a Giffen good?

Solution: The **income effect** must dominate the **substitution effect**.

In this case it is plausible because:

- Huge budget share (80%) → very strong income effect
- Few realistic alternatives → limited substitution effect

Overall Summary

Part 1: Demand Function

- Marshallian demand: $x^D(p_x, p_y, M)$
- Demand curve: graph of p_x vs x (typically decreasing)
- Properties: homogeneity of degree zero, negative slope

Part 2: Engel Curves

- Income-quantity relationship: $x^D(M)$
- Normal vs inferior goods
- Luxuries vs necessities
- Engel's law (food shares)

Part 3: Price and Income Effects

- Decomposition: substitution effect + income effect
- Price-consumption curve
- Giffen goods (a rare paradox)

Skills Acquired

- ✓ Derive the demand function from the optimal choice
- ✓ Graph the demand curve
- ✓ Distinguish normal/inferior/luxury/necessity goods
- ✓ Interpret Engel curves
- ✓ Understand the decomposition of the price effect
- ✓ Recognize when the Giffen paradox can arise

Optional Homework Exercises

1. $U = x^{0.4}y^{0.6}$. Derive $x^D(p_x, p_y, M)$ and then compute x^D for $p_x = 5$, $p_y = 10$, $M = 200$.
2. With demand $x^D = 80/p_x$, compute the quantity for $p_x = 2, 4, 8, 16$. Graph it.
3. $x^D = 0.2M/p_x$. Is pizza normal, inferior, a luxury, or a necessity? Justify.
4. $x^D = 50 - 0.01M$. What type of good is it? Draw the Engel curve.
5. Explain in your own words why a Giffen good requires three simultaneous conditions: (i) inferior, (ii) large budget share, (iii) dominant income effect.